

Improving Precision in Stratified Experiments: Design-Based Inference Meets ANCOVA

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Two sources of precision gains

Suppose we have a stratified random experiment, interested in SATE.

1. Design-based inference yields precision gains over EHW (robust) SEs (Athey & Imbens, 2017; Abadie et al., 2020)
 - Cluster-level treatment heterogeneity cancels out of within-strata Neyman variance
 - But typically only applied to post-treatment outcomes or differences
2. ANCOVA yields efficiency gains over post/DiD estimator (Frison & Pocock, 1992; McKenzie, 2012)
 - Efficiency maximized by conditioning on baseline outcome adjusted by ρ
 - Almost exclusively applied in model-based setups

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1. Can these two sources of precision gains be combined?
 - Yes, with a simple hybrid estimator
2. How big are the gains?
 - **Simulations:** Up to 39% lower s.e.s (equivalent to $2.7\times$ the effective sample size) relative to ANCOVA-EHW when cluster-level TE heterogeneity is large; negligible cost when it is not
 - **Real data:** Small gains from existing RCTs: few examples of high σ_τ and ρ simultaneously
3. Implications for experimental design:
 - Strengthens the argument for stratifying by tuples over pairs (Athey & Imbens, 2017) ($1.2\text{--}1.9\times$ smaller SEs under large heterogeneity, increasing in ρ)

Framework

Setup

- Population N , equal-sized clusters $g = 1, \dots, G$
- Individual randomization within cluster (2T/2C)
- baseline Y^0 and post outcome Y^1
- cluster-average effects τ_g , SATE $\bar{\tau} \equiv \frac{1}{G} \sum_g \tau_g$.
- Let $\sigma_\tau^2 \equiv \frac{1}{G} \sum_g (\tau_g - \bar{\tau})^2$, the *between-cluster* dispersion of treatment effects
- Let $\sigma_\varepsilon^2 \equiv \text{Var}(\varepsilon_{ig})$.

ANCOVA and Design-Based Estimators

ANCOVA

$$Y_{ig}^1 = \tau D_{ig} + \rho Y_{ig}^0 + \delta_g + \varepsilon_{ig}$$

Typically estimated with OLS, used with EHW (heteroskedasticity-robust) s.e.

$$\widehat{\mathbb{V}}_{\text{EHW}} \xrightarrow{p} \underbrace{\frac{4\sigma_\varepsilon^2}{N}}_{\text{true SATE variance}} + \underbrace{\frac{\sigma_\tau^2}{N}}_{\text{not relevant for SATE under balance}}$$

Design-based (levels)

$$\hat{\tau} = \frac{1}{G} \sum_{g=1}^G \hat{\tau}_g, \quad \hat{\tau}_g \equiv \bar{Y}_{Tg}^1 - \bar{Y}_{Cg}^1, \quad \mathbb{V}_{\text{Des}} = \frac{1}{G} \sum_{g=1}^G \left(\frac{s_{T,g}^2}{N_{Tg}} + \frac{s_{C,g}^2}{N_{Cg}} \right)$$

σ_τ^2 never enters $\widehat{\mathbb{V}}_{\text{Des}}$

» derivation

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Design-based (differences)

$$\hat{\tau} = \frac{1}{G} \sum_{g=1}^G \hat{\tau}_g, \quad \hat{\tau}_g \equiv (\bar{Y}_{Tg}^1 - \bar{Y}_{Cg}^1) - (\bar{Y}_{Tg}^0 - \bar{Y}_{Cg}^0), \quad \mathbb{V}_{\text{Des}} = \frac{1}{G} \sum_{g=1}^G \left(\frac{s_{T,g}^2}{N_{Tg}} + \frac{s_{C,g}^2}{N_{Cg}} \right)$$

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Design-based (hybrid)

$$\hat{\tau} = \frac{1}{G} \sum_{g=1}^G \hat{\tau}_g, \quad \hat{\tau}_g \equiv (\bar{Y}_{Tg}^1 - \bar{Y}_{Cg}^1) - \hat{\rho} (\bar{Y}_{Tg}^0 - \bar{Y}_{Cg}^0), \quad \mathbb{V}_{\text{Des}} = \frac{1}{G} \sum_{g=1}^G \left(\frac{s_{T,g}^2}{N_{Tg}} + \frac{s_{C,g}^2}{N_{Cg}} \right)$$

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►► derivation

The hybrid estimator

$$\hat{\tau}_g = (\bar{Y}_{Tg}^1 - \bar{Y}_{Cg}^1) - \hat{\rho}(\bar{Y}_{Tg}^0 - \bar{Y}_{Cg}^0), \quad \mathbb{V}_{\text{Des}} = \frac{1}{G} \sum_{g=1}^G \left(\frac{s_{T,g}^2}{N_{Tg}} + \frac{s_{C,g}^2}{N_{Cg}} \right)$$

Preferred approach is to plug in a cross-fitted $\hat{\rho}$

- Plug-in estimator in principle similar to Roth and Sant'Anna (2023) but cross-fitting can perform better with small G .
 - Nevertheless, cross-fitting introduces a small amount of noise $O(1/G)$ to s_T^2 and s_C^2
- Estimating ρ in-sample risks-overfitting (Freedman, 2008) or attributes σ_τ Lin (2013)
- In practice, also possible to estimate heterogeneous ρ (e.g. by arm, strata, or covariates)

► algorithm ► plug-in vs cross-fit

Simulation Evidence

Simulation design

DGP: $Y_{igt} = \alpha_t + \delta_g + \tau_g D_i \times 1t > 0 + \varepsilon_{igt}$

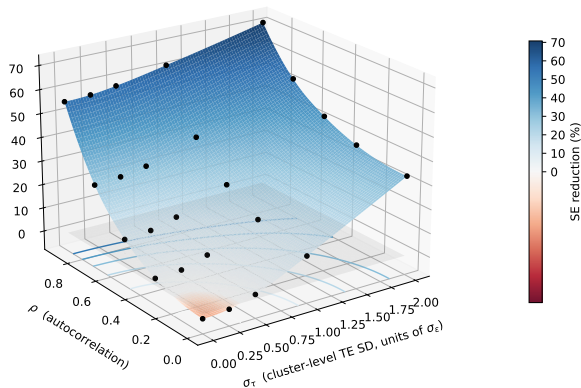
- 1 baseline, 1 endline, AR(1) errors with parameter ρ
 - $T \in \{3, 5\}$ multi-round variants in appendix
- $G = 50$ clusters, $N_g = 4$, 2T/2C per cluster; 1,000 reps per cell
- $\tau_g = \tau_0 + \sigma_\tau z_g$ fixed across reps (SATE)

Compare with grid over $\rho \in [0, 0.9]$, $\sigma_\tau \in [0, 2]$

1. ANCOVA with EHW s.e.
2. Neyman in levels when $\rho \leq .5$, differences otherwise, no ANCOVA adjustment
3. Hybrid

Hybrid gains over doing neither ANCOVA nor design

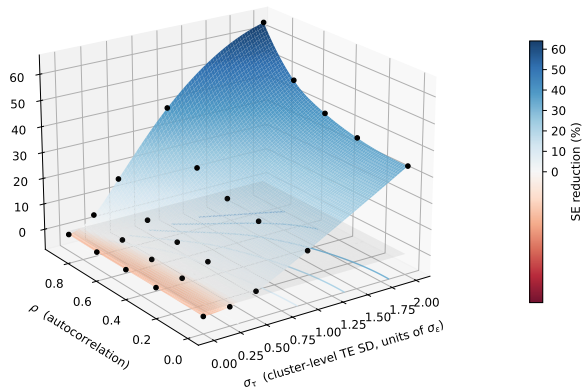
Hybrid vs doing neither (levels + HC1: no baseline, no design variance)
deterministic τ_g (SATE), $G = 50$, $N_g = 4$, 1000 reps/cell



12–61% smaller s.e. at $\sigma_\tau = \sigma_\epsilon$ ($1.3\text{--}6.7\times$ the effective sample)

Hybrid vs ANCOVA-EHW: gains increase in σ_τ and ρ

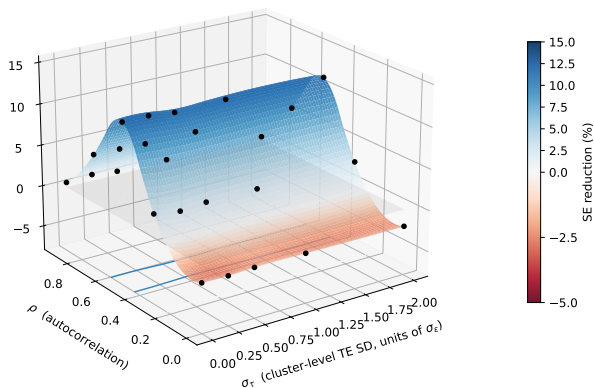
Hybrid vs ANCOVA-HC1 (common practice under individual randomization)
deterministic τ_g (SATE), $G = 50$, $N_g = 4$, 1000 reps/cell



12–39% smaller s.e. at $\sigma_\tau = \sigma_\epsilon$ ($1.3\text{--}2.7\times$ the effective sample)

Hybrid vs best unadjusted design-based

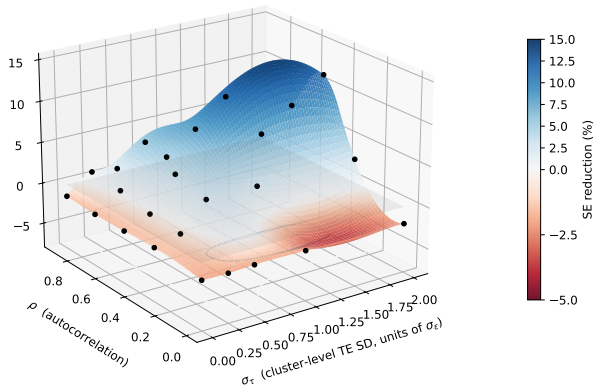
Hybrid vs best of $\{AI(\Delta y), AI(levels)\}$ (design-based, no ANCOVA)
deterministic τ_g (SATE), $G = 50$, $N_g = 4$, 1000 reps/cell



Best non-ANCOVA design-based is post estimator at $\rho \leq .5$, DiD at $\rho \geq .5$ (McKenzie, 2012).
Pure ρ channel, flat in σ_τ : up to 12% at the $\rho = 0.5$ crossover (+28% effective sample)

Hybrid vs the best feasible alternative

Hybrid vs next-best of $\{AI(\Delta y), AI(levels), ANCOVA-HC1\}$
deterministic τ_g (SATE), $G = 50$, $N_g = 4$, 1000 reps/cell



Peak 11.5% (+28% effective sample)

Evidence from Real Experiments

Applying the hybrid to 11 published RCTs

Standard error reductions

application	$\hat{\rho}$	vs neither	vs ANCOVA-HC	vs best design	rule vs HC
DKR 2011 (Kenya fertilizer)	0.28	+3%	+2.4%	-1.3%	+2.4%
Beaman 2013 (Mali fertilizer)	0.47	+8%	-1.7%	+3.6%	-1.7%
Riley 2024 (Uganda loans)	0.23	+12%	+1.8%	+8.8%	+1.8%
McKenzie-Puerto 2021 (Kenya training)	0.52	+5%	-0.3%	+3.7%	-0.3%
Hussam et al. 2022 (India grants)	0.33	+15%	-2.5%	+15.2%	-2.5%
Mbiti et al. 2019 (Tanzania schools)	0.97	+42%	-1.3%	-3.1%	+1.8%
Muralidharan et al. 2016 (Smartcards)	0.03	+1%	+2.0%	-1.5%	+3.5%
Bandiera et al. 2020 (ELA Uganda)	0.17	+3%	+3.9%	-3.5%	+3.9%
Banerji et al. 2017 (mother literacy)	1.00	+79%	+4.8%	-0.6%	+5.4%
King et al. 2009 (Seguro Popular)	0.10	+13%	+1.1%	+0.2%	+0.8%
Ligon and Silver (2026) (PHLs)	0.04	+3%	+2.7%	-4.1%	+6.5%

- Largest gains where baselines are informative (test scores, $\rho \rightarrow 1$)
- “rule”—just use design-based if $\rho > .85$ or $\rho < .15$ (estimable from data)

(Duflo et al., 2011; Beaman et al., 2013; Riley, 2024; McKenzie & Puerto, 2021; Hussam et al., 2022; Mbiti et al., 2019; Muralidharan et al., 2016; Bandiera et al., 2020; Banerji et al., 2017; King et al., 2009; Ligon & Silver, 2025)

» diagnosing σ_{τ}

Design Implications

Tuples over Pairs

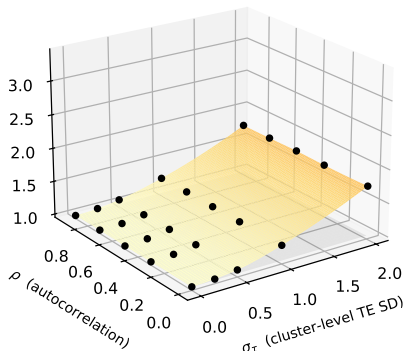
Hybrid estimator amplifies the case for fine stratification with at least 2 units per arm per cluster (Athey & Imbens, 2017)

1. Pairs give up the within-stratum variance that design-based inference uses (Athey & Imbens, 2017)
 - Between-pair spread carries σ_τ^2 and noise inseparably (de Chaisemartin & Ramirez-Cuellar, 2024)
2. Covariate adjustment (including ANCOVA) amplifies the penalty
 - SE ratio 1.00 at $\sigma_\tau = 0$; 1.2–1.9 at $\sigma_\tau = \sigma_\varepsilon$; largest at high ρ
3. Shrinks the worst-case variance inflation factor under bounded outcomes relative to pairs $\propto \frac{N_g}{N_g - 1}$ (Hull, 2026)

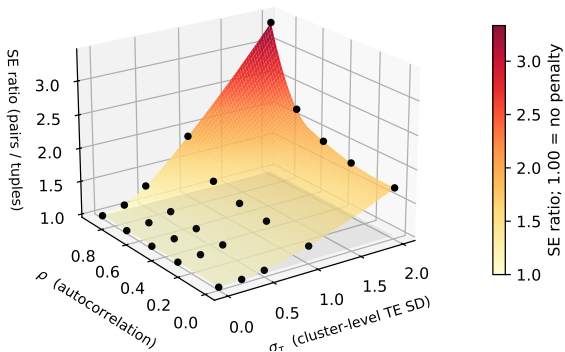
The pairs penalty over (ρ, σ_τ)

The pairs penalty: a σ_τ main effect that adjustment amplifies through ρ (headline grid, $G=50$, 2T/2C, 1000 reps)

Unadjusted: pairs-dCRC / tuples-AI (levels)
penalty is pure σ_τ — flat in ρ



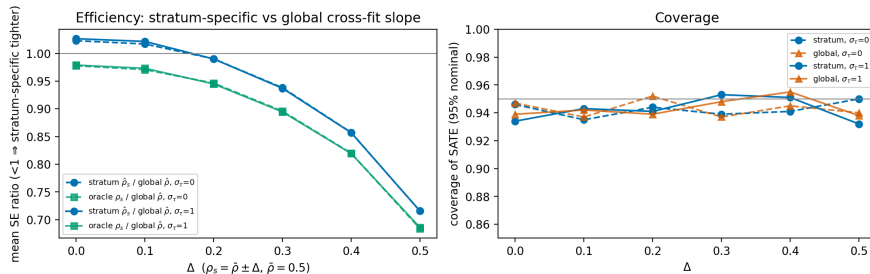
Adjusted (same cross-fit $\tilde{Y}$): pairs / tuples
 ρ amplifies the penalty



Estimate ρ at the resolution of your strata

ρ may also exhibit heterogeneity across strata. Can estimate it heterogeneously by (super-)strata or covariates:

Stratum-specific ρ in the cross-fitted hybrid — S=2 strata $\times$ 24 tuples, controls-only, min 20 OOF controls, 1000 reps



- Cost when slopes are equal: $\approx 2\%$; 28% tighter at $\Delta = 0.5$; coverage remains correct
- Heterogeneity in ρ detectable from control data ex ante

[» the recipe](#)

Takeaways

Combining design-based inference with ANCOVA yields precision gains for SATE under cluster-level treatment effect heterogeneity

- Comes at a small precision cost if treatment effects don't vary at the cluster level
- Gains are modest but mostly positive in existing RCT data, but simulations show when payoffs can be large

Implications for Design

- Strengthens the Athey and Imbens (2017) case for fine tuple-level stratification (as opposed to pairs)

Implications for Analysis

- Worth pre-registering as primary when ρ and σ_τ expected to be high
- Even without strong priors, pre-specify contingent on triggers for $\hat{\rho}$ and differences in TEs across strata.

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Appendix

Derivation: where σ_τ^2 enters the EHW variance

Regression fits a common τ , so $\tilde{\tau}_g \equiv \tau_g - \bar{\tau}$ lands in the residual:

$$\hat{u}_i = \hat{\varepsilon}_i + \tilde{\tau}_{g(i)} \ddot{D}_i, \quad \ddot{D}_i = D_i - \bar{D}_g = \pm \frac{1}{2} \Rightarrow \sum_i \ddot{D}_i^2 = \frac{N}{4}$$

Meat of the sandwich, term by term: $\mathbb{E}[\ddot{D}^2 \hat{\varepsilon}^2] = \frac{\sigma_\varepsilon^2}{4}$, cross term 0, $\mathbb{E}[\ddot{D}^4 \tilde{\tau}_g^2] = \frac{\sigma_\tau^2}{16}$:

$$\hat{\mathbb{V}}_{\text{EHW}} = \frac{\sum_i \ddot{D}_i^2 \hat{u}_i^2}{(\sum_i \ddot{D}_i^2)^2} \xrightarrow{p} \frac{N(\frac{\sigma_\varepsilon^2}{4} + \frac{\sigma_\tau^2}{16})}{(N/4)^2} = \frac{4\sigma_\varepsilon^2 + \sigma_\tau^2}{N}$$

True randomization variance: the within-cluster contrast is $\tau_g + \nu_g$ with τ_g fixed under the design $\Rightarrow V_{\text{true}} = 4\sigma_\varepsilon^2/N$

- Conservativeness ratio $1 + \sigma_\tau^2/(4\sigma_\varepsilon^2)$: grows with ρ once adjustment shrinks σ_ε^2

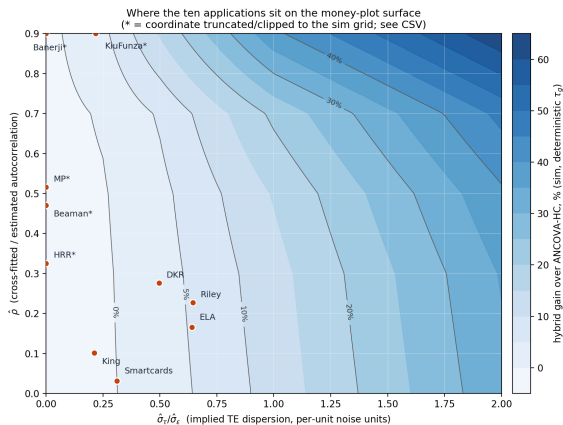
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Stratum-specific ρ : the process

1. **Choose strata:** any pre-treatment partition with multiple tuples per cell – randomization strata, survey wave, geography, frame
2. **Diagnose ex ante:** fit control-only within-tuple slopes $\hat{\rho}_s$ per stratum; compare their weighted spread to its own sampling noise (excess variance = $\text{Var}_w(\hat{\rho}_s) - \overline{SE^2}$) – positive and $\Delta \gtrsim 0.2$ says proceed
3. **Fit:** cross-fit $K = 2$ at the tuple level; $\hat{\rho}_s$ from out-of-fold controls pooled within the stratum; *pre-specified* fallback to the pooled slope in any (fold $\times$ stratum) cell with < 20 out-of-fold controls
4. **Adjust and infer:** $\tilde{Y}_i = \bar{y}_{post,i} - \hat{\rho}_{s(i)} y_{0,i}$, stratified AI on tuples – stratum membership is pre-treatment, so unbiasedness and the design-based variance are untouched
 - Gain formula: assignment-weighted between-stratum variance of the slopes; cost $O(S/n)$

Diagnosing σ_τ : the cluster/EHW fingerprint

The observable fingerprint of σ_τ is the cluster/EHW SE ratio (sim: 1.00 at $\sigma_\tau = 0$, 1.27 at $\sigma_\tau = \sigma_\varepsilon$, 1.51 at $2\sigma_\varepsilon$); estimates for all ten replication studies:



- Every study sits at $\hat{\sigma}_\tau / \hat{\sigma}_\varepsilon \leq 0.65$ — the low-gain edge; mapping depends on (G, N_g) , treat as anchor

Plug-in vs cross-fit $\hat{\rho}$, by number of clusters

Same estimator and stratified Neyman variance; only the slope fit differs ($\sigma_\tau = 0$, $\rho = 0.5$, 2{,}000 reps; $\sigma_\tau = 1$ similar)

G	coverage: plug-in	cross-fit	SE/sd: plug-in	cross-fit
16	0.920	0.943	0.91	0.98
50	0.949	0.954	0.98	1.00
200	0.947	0.948	0.98	0.99

- In-sample fitting flatters the residuals: reported SE understates the spread at small G
- The plug-in point estimator is slightly *tighter* — the gap is validity, not precision
- By $G \approx 50$ the two coincide: cross-fitting is insurance for the few-cluster regime

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Hybrid: algorithm and cross-fitting details

1. Split clusters into K folds
2. For fold k and arm $a \in \{T, C\}$, from *out-of-fold units of arm a* , with $\ddot{y}$ denoting within-cluster demeaning:

$$\hat{\rho}_{a,k} = \frac{\sum_{i: D_i=a, k(g_i) \neq k} \ddot{y}_{0,i} \ddot{y}_{post,i}}{\sum_{i: D_i=a, k(g_i) \neq k} \ddot{y}_{0,i}^2}$$

1. Adjust on the centered baseline: $\tilde{Y}_i = \bar{y}_{post,i} - \hat{\rho}_{a(i), k(g_i)} (y_{0,i} - \bar{y}_{0,g(i)})$
2. Stratified AI on $\tilde{Y}$, cluster = stratum
 - Arm-specific slopes = the Lin (2013) / Roth-Sant'Anna (2023) efficient-coefficient target; cross-fitting upgrades their asymptotic plug-in guarantee to exact finite- G unbiasedness
 - Centering by the *fixed, pre-treatment* cluster mean $\bar{y}_{0,g}$ keeps the contrast unbiased when $\hat{\rho}_T \neq \hat{\rho}_C$
 - Cluster-level folds preserve $\geq 2T/2C$ per stratum and make the slope error stratum-constant — it enters $s_{a,g}^2$ as a coefficient shift, priced by $\hat{V}$